On the Interpretability of Diffusion-Based Recommendation Systems

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Abstract

Diffusion models have recently been adapted for recommendation tasks, framing item ranking as a denoising process over user-item interaction embeddings. While these models report strong offline accuracy, their iterative, stochastic generation process makes it difficult to explain any individual recommendation to end users or auditors. We propose a post-hoc attribution method that traces a recommended item back through the reverse diffusion trajectory to identify which historical interactions most influenced the final ranking. Across two public benchmark datasets, our method produces explanations that are rated as more helpful than baseline attention-weight explanations in a small user study, while adding negligible inference overhead.

1. Introduction

Recommendation systems increasingly sit behind consequential decisions — what job listing is shown first, which product is promoted, what article appears at the top of a feed. As diffusion-based architectures achieve state-of-the-art offline metrics on recommendation benchmarks, the field faces a familiar tension: model capability is outpacing model interpretability.

Unlike attention-based recommenders, where attribution can often be read directly from attention weights, diffusion recommenders generate a ranking through many small denoising steps, none of which individually corresponds to an interpretable decision.

2. Proposed Method

We introduce trajectory attribution: at each reverse diffusion step, we measure the gradient of the emerging item embedding with respect to each historical interaction in the user's profile, then aggregate these gradients across steps into a single per-interaction attribution score. The top-scoring interactions are surfaced as the explanation for a given recommendation.

3. Evaluation

We evaluate on two public interaction datasets and run a 40-participant user study comparing our trajectory attributions against attention-weight baselines and a random-interaction control. Participants rated trajectory attributions as more helpful on a 5-point Likert scale, though the gap narrows for users with very short interaction histories.

4. Limitations

Our attribution method assumes access to gradients at each diffusion step, which increases inference cost by roughly 18% compared to standard sampling; batching mitigates but does not eliminate this overhead.

References

[1] Fictional citation for demonstration purposes only.
[2] This document was generated as sample content for a MediaPipe sample app.